Conversion of units

12 in = 1 ft
3 ft = 1 yd
1760 yds = 1 mi
5280 ft = 1 mi

10 mm = 1 cm
100 cm = 1 m
1000m = 1 km
(basic units are m or km)

(English-Metric conversions: 1 inch = 2.54 cm; 1 mile = 1.61 km)

Examples:

4000 in = ? ft

Notice that 1 ft = 12 in. We write our "1" as $\frac{1ft}{12in}$ so that "in" in 4000 in will cancel.

$4,000in = \frac{4,000in}{1} \cdot \frac{1ft}{12in} = \frac{4,000}{12} ft \approx 333.33 ft$. notice that the "in" canceled

18 ft = ? in

$18ft = \frac{18ft}{1} \cdot \frac{12in}{1ft} = 216in$

Notice that 1 ft = 12 in. This time we write our "1" as 12in / 1ft so that the "ft" cancel leaving only "in" as the unit.

2.3 mi = ? in

$2.3mi = \frac{2.3mi}{1} \frac{5280ft}{1mi} \frac{12in}{1ft} = \frac{(2.3)(5280)(12)}{1} in = 145728in$

notice that the "mi" and the"ft" cancel

146000 cm = ? km

Just like in the previous example we don't have a direct conversion between centimeters (cm) and kilometers (km). However, both are connected through meters (m). Therefore, we will use the "1" in two different ways: $1 = \frac{1m}{100cm}$ and $1 = \frac{1km}{1000m}$. Using both, we get the following:

$146000cm = \frac{146000cm}{1} \frac{1m}{100cm} \frac{1km}{1000m} = \frac{146000}{(100)(1000)} km = 1.46km$

notice that the "cm" and the "m" cancel

1,000,000 mm^2 = ? km^2

This time we will use two conversions 1 m = 1000 mm and 1 km = 1000 m. Each of these two conversion will be used twice, since we are using square units:

$1000000mm^2 = \frac{1000000mm^2}{1} \cdot \frac{1m}{1000mm} \cdot \frac{1m}{1000mm} = 1m^2$ Then we continue:

$1m^2 = 1m^2 \cdot \frac{1km}{1000m} \cdot \frac{1km}{1000m} = 0.000001km^2$. Therefore $1000000mm^2 = 0.000001km^2$.

notice that "mm^2" cancels with "mm mm" in the first row, and "m^2" cancels with "m m" in the second row of calculations above.

If a car is traveling 90 mi/hr, how many feet will the car travel in 1 sec? In 5 sec?

A direct way to convert these units is:

$$90mi/hr = \frac{90mi}{1hr}\frac{5280ft}{1mi}\frac{1hr}{60\min}\frac{1\min}{60\sec} = \frac{(90)(5280)}{(60)(60)}ft/\sec = 132ft/\sec$$

Alternatively, we can derive our own conversion formulas separately for miles and for hours: $1mi = 5280ft$ and $1hr = 60min = 60 \cdot 60sec = 3600sec$. We then replace miles with 5280feet and hours with 3600sec. Therefore $\frac{90mi}{hr} = \frac{90 \cdot 5280ft}{3600sec} = 132ft/sec$

In 1 second the car will travel 132 ft. The car will travel (5 sec)(132 ft/sec) = 660 ft. in 5 seconds.

Dimensional Analysis

Dimension	Symbol
Length	L
Time	T
Mass	M

Examples:

Consider the equation is dimensionally correct, what is the dimension of k?

$$y = x + \frac{1}{2}kx^3$$

If in the above equation x and y were both lengths (dimension L) and 1/2 is a dimensionless number, then for the $\frac{1}{2}kx^3$ term to have the same dimension as the other two, we would need:

$$\text{dimension of } k \times \mathrm{L}^3 = \mathrm{L}$$

$$\therefore \text{dimension of } k = \frac{\mathrm{L}}{\mathrm{L}^3} = \mathrm{L}^{-2}$$

So k would have dimensions of one over area, i.e., $[k] = \mathrm{L}^{-2}$.

Example The expressions for **kinetic energy** $E = \frac{1}{2}mv^2$ (where m is the mass of the body and v is its speed) and **potential energy** $E = mgh$ (where g is the acceleration due to gravity and h is the height of the body) look very different but both describe energy. One way to see this is to note that they have the same dimension.

Dimension of **kinetic energy**

$$\tfrac{1}{2}mv^2 \Rightarrow \mathrm{M}(\mathrm{L}\,\mathrm{T}^{-1})^2 = \mathrm{M}\,\mathrm{L}^2\,\mathrm{T}^{-2}$$

Dimension of **potential energy**

$$mgh \Rightarrow \mathrm{M}(\mathrm{L}\,\mathrm{T}^{-2})\,\mathrm{L} = \mathrm{M}\,\mathrm{L}^2\,\mathrm{T}^{-2}$$

Both expressions have the same dimensions, they can therefore be added and subtracted from each other.

$$\begin{aligned} \text{Dimension of volume} &= \mathrm{L} \times \mathrm{L} \times \mathrm{L} \\ &= \mathrm{L}^3 \end{aligned}$$

$$\begin{aligned} \text{Dimensions of speed} &= \frac{\mathrm{L}}{\mathrm{T}} \\ &= \mathrm{L}\,\mathrm{T}^{-1} \end{aligned}$$

$$\begin{aligned} \text{Dimensions of acceleration} &= \frac{\mathrm{L}\,\mathrm{T}^{-1}}{\mathrm{T}} \\ &= \mathrm{L}\,\mathrm{T}^{-2} \end{aligned}$$

Quiz Hooke's law states that the force, F, in a spring extended by a length x is given by $F = -kx$. From Newton's second law $F = ma$, where m is the mass and a is the acceleration, calculate the dimension of the spring constant k.

(a) $\mathrm{M}\,\mathrm{T}^{-2}$ (b) $\mathrm{M}\,\mathrm{T}^{2}$ (c) $\mathrm{M}\,\mathrm{L}^{-2}\,\mathrm{T}^{-2}$ (d) $\mathrm{M}\,\mathrm{L}^{2}\,\mathrm{T}^{2}$

Solution to Quiz: From Hooke's law, $F = -kx$, we see that we can write

$$k = \frac{F}{x}$$

Now $F = ma$, so the dimensions of force are given by

$$\begin{aligned} [\mathrm{F}] &= \mathrm{M} \times (\mathrm{L}\,\mathrm{T}^{-2}) \\ &= \mathrm{M}\,\mathrm{L}\,\mathrm{T}^{-2} \end{aligned}$$

Therefore the spring constant has dimensions

$$\begin{aligned} [k] &= \frac{\mathrm{M}\,\mathrm{L}\,\mathrm{T}^{-2}}{\mathrm{L}} \\ &= \mathrm{M}\,\mathrm{T}^{-2} \end{aligned}$$